Star NX-1020 — Resident Font Catalog

Five resident typefaces recovered from the printer's font ROM, with bitmaps and metadata. · Interactive ROM map ▸

Source: **ROM1** (IC7, D27C1001, 128 KB font/data ROM) · firmware ROM2 v3.2 (32 KB) · CPU Mitsubishi M50734

> The NX-1020 is a 9-pin impact dot-matrix printer. Its resident fonts are **fixed-size bitmaps** stored in a 128 KB ROM, read through a gate array via a banked window. Each face has an 8-byte-per-character **metrics table** (in ROM bank 2) pointing at the packed glyph bitmaps (in banks 0/3). Draft glyphs are 8 dots tall (single pass); the four near-letter-quality faces are 16 dots tall (two half-dot-offset passes, planar-interleaved). All offsets below are file offsets into the ROM1 image.

SUMMARY

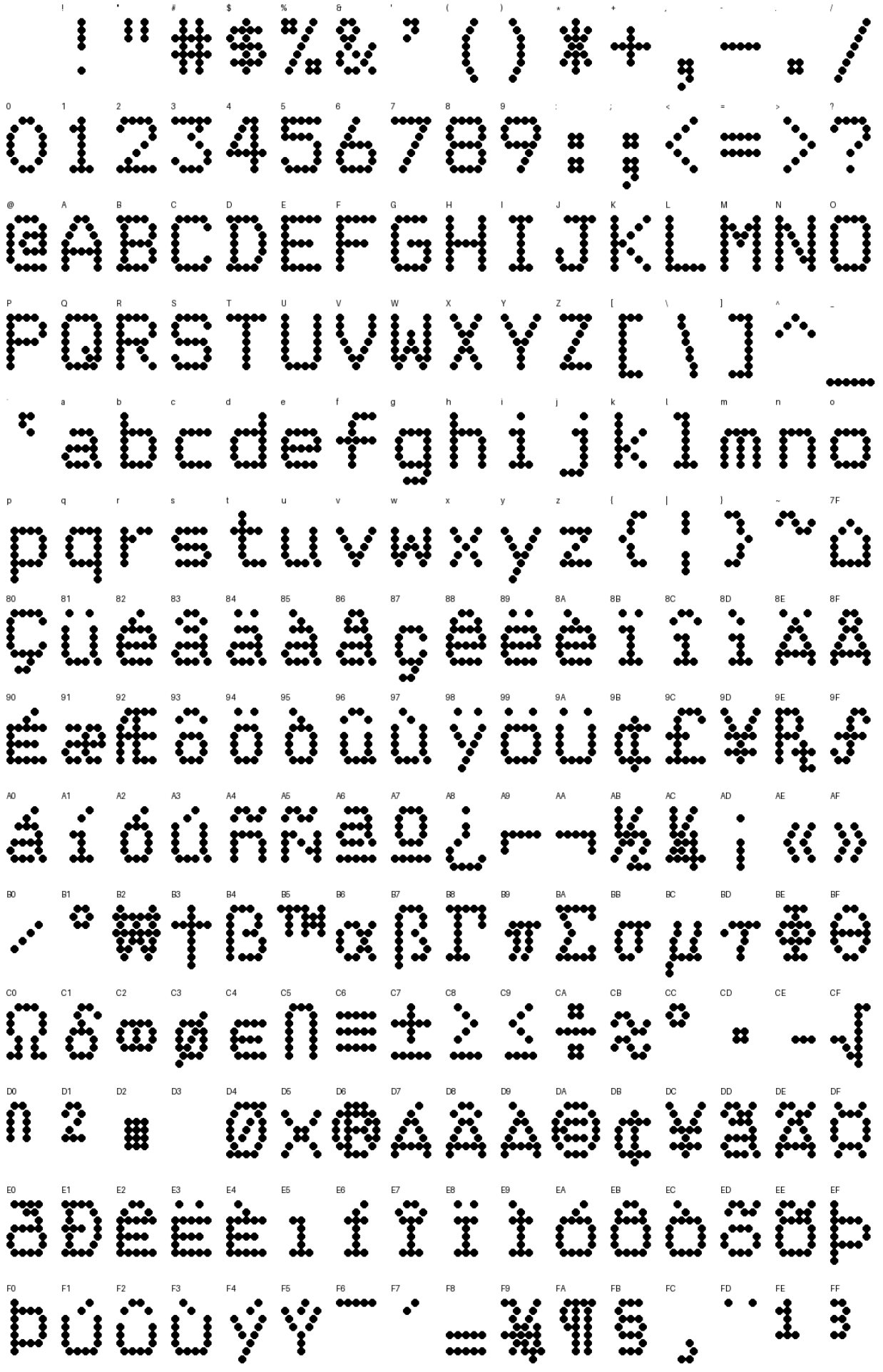
FONT	TYPE	ROM BANK	METRICS @	GLYPH DATA @	GLYPHS	SELECT
Draft	8-dot draft	ROM bank 3	0x132C1	0x18AA2	224	ESC x 0 / EDS
Courier	16-dot NLQ · serif	ROM bank 3	0x14F21	0x1B917	255	ESC k 0
Sanserif	16-dot NLQ · sans	ROM bank 3	0x15C81	0x1DDE3	228	ESC k 1
Orator	16-dot NLQ · bold caps	ROM bank 3 → 0 (wraps)	0x168F1	0x1FBEB → 0x00000	228	ESC k 7
Script	16-dot NLQ · cursive	ROM bank 0	0x17561	0x01CBC	228	ESC k 4

Draft 8-dot draft

The everyday face (this card = the Pica set). 8-dot single-pass, column-major (1 byte/column, bit 7 = top pin), proportional width, 150 cps. Elite and the two separately stored High-Speed Draft faces (200/225 cps) are in the sheets below. Selected by `ESC x 0` or the EDS print-mode switch.

Format	8-dot single-pass, 1 byte/col, proportional	ROM bank	ROM bank 3
Metrics table	0x132C1 (bank 2)	Glyph data	0x18AA2
Glyph count	224	Bitmap data	~2.5 KB
Descriptor idx	—	Selection	ESC x 0 / EDS
Native size	≈ 9 pt	Print speed	150–225 cps

NX-1020 draft - 224 glyphs (each shown in its 6-box cell)



Courier

16-dot NLQ · serif

The default near-letter-quality serif face. 16-dot, two-pass planar-interleaved (even rows from the first pass, odd rows from the half-dot-offset second pass). Full character set: IBM graphic symbols, ASCII, accented/international, box-drawing and Greek. Selected by `ESC k 0`; internal descriptor index 0x14.

Format	16-dot NLQ, planar-interleaved (2 bytes/col)	ROM bank	ROM bank 3
Metrics table	0x14F21 (bank 2)	Glyph data	0x1B917
Glyph count	255	Bitmap data	9.4 KB
Descriptor idx	0x14	Selection	ESC k 0
Native size	≈ 12 pt	Print speed	37.5–45 cps

NX-1020 courier - 255 glyphs (each shown in its 6-box cell)



Sanserif

16-dot NLQ · sans

Near-letter-quality sans-serif face. Same 16-dot planar format as Courier and the same full character set. Shares the IBM graphic-symbol glyphs with Courier (identical pointers), so its own table begins at the letters. `ESC k 1`, index 0x1e.

Format	16-dot NLQ, planar-interleaved (2 bytes/col)	ROM bank	ROM bank 3
Metrics table	0x15C81 (bank 2)	Glyph data	0x1DDE3
Glyph count	228	Bitmap data	7.7 KB
Descriptor idx	0x1e	Selection	ESC k 1
Native size	≈ 12 pt	Print speed	37.5–45 cps

NX-1020 sanserif - 228 glyphs (each shown in its6-box cell)



Orator

16-dot NLQ · bold caps

A heavy display face for legibility at a distance — the lowercase keys print as small capitals (Orator's signature). Its glyph data lives in ROM bank 0, and the alphabet straddles the bank 3→bank 0 seam (the boundary 'B' glyph needed special handling). `ESC k 7`, index 0x28.

Format	16-dot NLQ, planar-interleaved (2 bytes/col)	ROM bank	ROM bank 3 → 0 (wraps)
Metrics table	0x168F1 (bank 2)	Glyph data	0x1FBEB → 0x00000
Glyph count	228	Bitmap data	8.4 KB
Descriptor idx	0x28	Selection	ESC k 7
Native size	≈ 12 pt	Print speed	37.5–45 cps

NJX-1020 orator - 228 glyphs (each shown in its 6-box cell)



Script 16-dot NLQ · cursive

The one inherently cursive resident design — connected, flowing letterforms (elsewhere 'italic' is just a programmatic slant). Glyph data is in ROM bank 0. `ESC k 4`, index 0x32.

Format	16-dot NLQ, planar-interleaved (2 bytes/col)	ROM bank	ROM bank 0
Metrics table	0x17561 (bank 2)	Glyph data	0x01CBC
Glyph count	228	Bitmap data	8.5 KB
Descriptor idx	0x32	Selection	ESC k 4
Native size	≈ 12 pt	Print speed	37.5–45 cps

NJX-1020 script - 228 glyphs (each shown in its 6-box cell)



ADDITIONAL 8-DOT SETS — DRAFT PITCHES & EXTENDED CHARACTERS

The **Draft** card above is Pica (10 CPI, 9×11). The 8-dot region of the ROM is a run of complete CP437 Draft **faces** at different pitches (each carries the full alphabet + accented/extended characters), interleaved with shared **companion** sets (accented upper-half and box-graphics) that have *no alphabet of their own*. Each sheet below is bounded to one set. (Bold/double-strike/expanded etc. remain *programmatic*, not stored.)

Draft — Elite (12 CPI)

8-dot · ALPHABET face · max 9 cols · group 0, elite slot

The Elite (12 cpi) Draft face: the same 8-dot format redrawn in 9-column cells. Selected by `ESC M` in draft quality (font index +1).

Metrics table	`0x12B91 (bank 2)`
Format	`8-dot, 1 byte/col`
Glyphs	`179`

Draft 8-dot, 9-wide @0x12B91 (identity undetermined - one plausibly Elite; NOT HS-Draft)



High-Speed Draft — Pica (10 CPI)

8-dot · ALPHABET face · max 9 cols · group 1, pica slot

High-Speed Draft's own stored pica face — drawn in 9 columns instead of Draft's 11, so each glyph needs fewer head fires: that (plus faster carriage timing) is where the 200/225 cps comes from, and why HS output looks thinner. Selected when the HS-active flag picks font group 1 (traced @0x86EC).

Metrics table	`0x14509 (bank 2)`
Format	`8-dot, 1 byte/col`
Glyphs	`179`

Draft 8-dot, 9-wide @0x14509 (identity undetermined - one plausibly Elite; NOT HS-Draft)



High-Speed Draft — Elite (12 CPI)

8-dot · ALPHABET face · max 7 cols · group 1, elite slot

High-Speed Draft's elite face, drawn in just 7 columns — the sparsest resident set. (Previously mislabeled 'Condensed': real condensed printing is a programmatic dot-pitch transform of the pica/elite cells, not a stored face.)

Metrics table	`0x13DE1 (bank 2)`	Glyphs	179
Format	`8-dot, 1 byte/col`		

Draft CONDENSED (17 CPI, w<=7) @0x13DE1

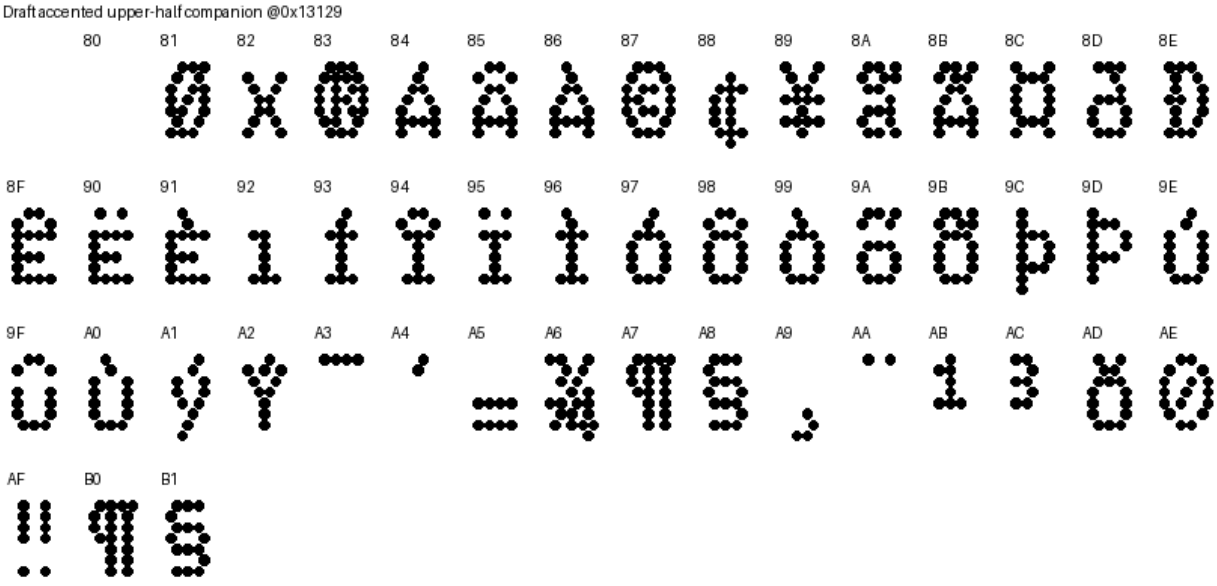


Extended — accent / upper-half companion

8-dot · companion · no alphabet

8-dot accented / international upper-half glyphs (Ç ü é â ä …) shared by the Draft faces. A companion set — it has no alphabet of its own.

Metrics table	`0x13129 (bank 2)`	Glyphs	`51`
Format	`8-dot, 1 byte/col`		



Extended — box graphics companion

8-dot · companion · no alphabet

8-dot box-drawing / block-graphics + accented upper-half (IBM line/shade chars). A companion set with NO alphabet of its own — the Draft alphabets are in the face sheets above. (Index-labeled, since these aren't a simple ASCII run.)

Metrics table	`0x13851 (bank 2)`
Format	`8-dot, 1 byte/col`
Glyphs	`178`

Draft box-graphics + upper-half companion (no alphabet; index-labeled) @0x13851



Format legend. 8-dot draft: column-major, 1 byte/column, bit 7 = top pin, proportional width. **16-dot NLQ**: planar-interleaved — first *h* bytes = even rows (0,2,…,14), next *h* = odd rows (1,3,…,15), where *h* = width/2 columns at half horizontal pitch. Width = pointer difference between consecutive characters.

Notes. Bold / condensed / double-width / double-strike / super-subscript and (for the upright faces) italic are produced *programmatically* by the firmware, not stored as separate glyphs — Script is the only inherently cursive design. Selection codes are the ESC/P `ESC k` typeface numbers; the internal descriptor indices differ.

Generated from `UPD27C1001A_DIP32_HN62321APS32_nx1020_ROM1.bin` by `build_presentation.py`.